Sahaj Bhadja

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Summary

Software engineer completing an MSCS at USC with 2+ years using **LLMs** and **prompt engineering** to refactor, document, and validate production code. Strong in **Python/Java**, shift-left testing, performance optimization, and human-in-the-loop AI-assisted SDLC workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Advanced NLP, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Data Structures, Algorithms, OOP, Software Engineering

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Refactored asynchronous **LLM** orchestration in **Python**/FastAPI using AI-assisted analysis, improving throughput by **3x** while reviewing generated changes for functional parity, security, and maintainability.
- Implemented human-in-the-loop validation and CI/CD releases for **5K+ monthly jobs**, reducing dashboard latency from **1.2s to 350ms** through algorithmic and query optimization across production systems.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **prompt engineering** workflows with **Claude** and GPT-4o for grading automation, refining prompts to improve consistency across **10K+ daily requests** while reviewing AI-generated outputs before release.
- Built modular Flask services with unit and integration tests, integrating AI tooling into GitHub Actions CI/CD and documenting API schemas for maintainable application modules.

USC AD Labs **Los Angeles, CA**
Software Engineer - ML Systems *January 2025 – May 2025*

- Maintained **Python** pipelines processing **100K+ trajectory logs**, refactoring data workflows for clarity and reproducibility while collaborating with senior researchers on performance improvements.

Aasma Technology Solutions **India**
Software Engineer - (SecureSync) *August 2023 – July 2024*

- Refactored legacy API modules with OOP design patterns, reducing technical debt by **20%** and response times by **30%** while preserving behavior through **20+ PyTest** regression suites.
- Integrated automated testing into Jenkins CI/CD across Linux, Windows, and macOS, supporting shift-left quality practices and detailed failure reporting for production validation workflows.

Aasma Technology Solutions **India**
Software Engineer Intern - (VibrantChem) *May 2023 – August 2023*

- Improved microservice maintainability and Docker/Jenkins automation, cutting deployment cycles by **50%** with documented APIs and structured logging for operations teams.

Skills

AI & SDLC: LLMs, Claude, GPT-4o, prompt engineering, generative AI, AI-assisted refactoring, technical documentation, human-in-the-loop review, legacy code analysis, clean code, design patterns, performance optimization
Engineering: Python, Java, C++, Go (familiar), data structures, algorithms, unit testing, integration testing, shift-left testing, PyTest, CI/CD, Jenkins, GitHub Actions, Git, Docker, Kubernetes (familiar), FastAPI, Flask, REST APIs, DevOps, Agile

Projects

Big-Brain Live | *LLMs, Prompt Engineering, Documentation*

- Built an AI document platform using **LLMs** over **50K+ records**, generating summaries and documentation with automated testing and iterative prompt refinement.

Chat2SQL Code | *Open Source, Refactoring, Docker, CI/CD*

- Extended a large open-source codebase with AI-assisted SQL workflows, refactoring Java/TypeScript modules and containerizing services with Docker and cross-platform pipelines.

CodeRefactor Lab Code | *Python, Claude, Unit Tests, Legacy Modernization*

- Modernized legacy **Python** modules with **Claude**-guided refactoring, validating functional parity through unit tests and auto-generated API documentation.